

CMAPSS Anomaly Detection Using VAEs

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Abstract

Generative AI techniques have been widely applied across various domains, and anomaly detection in turbofan engines is challenging due to gradual degradation and the absence of explicit anomaly labels. In this work, an empirical evaluation of Variational Autoencoder (VAE)-based anomaly detection on the NASA CMAPSS turbofan engine dataset is conducted, investigating four interconnected design choices: anomaly score function, input window size, regularization of β , and alert threshold strategy. We compare a non-temporal MLP-VAE and a temporal GRU-VAE, showing that GRU-VAE produces a more operating condition-agnostic latent representation. Experiments demonstrate that no single configuration universally dominates, with β showing the strongest influence on overall performance. Moreover, a model with a posterior collapse achieves the highest performance both for AUROC of 94% and F1 of 70%, suggesting that overly regularized latent representations unexpectedly act as a better anomaly signal for CMAPSS dataset. [GitHub Link](#)

1 Introduction

Anomaly detection for industrial settings is a challenging task for various reasons, and to detect anomalies in industrial systems, traditional approaches (Liu et al., 2016) monitor sensor data based on predefined thresholds. However, such approaches lack generalization ability where they struggle with out-of-distribution data, failing to detect anomalies in more general setups. Compared to this, data-driven anomaly detection methods leverage massive time-series data to generate more generalized anomaly thresholds (Zhang et al., 2020).

In this work, we conduct an empirical evaluation using generative techniques to detect anomalies in turbofan jet engine dataset, namely CMAPSS NASA (2025). Unlike other conventional anomaly detection problems, CMAPSS contains gradual degradation cycles of engine sensors which makes the problem challenging, since there are no specific anomalous points. To the best of my knowledge, anomaly detection for CMAPSS data is yet underexplored compared to other anomaly detection problems, due to the lack of explicit anomaly labels and the gradual degradation data.

Given the nature of the unsupervised learning, β -Variational Autoencoder (β -VAE) is used to systematically investigate four design choices that significantly affect model performance: an anomaly scoring function choice, the effect of temporal difference through input window size, the role of β affecting the overall performance, and alert threshold strategy.

2 Related Work

2.1 Anomaly Detection with Generative Models

β -VAEs (Burgess et al., 2018) aim to train encoder and decoder to regularize inputs to latent space then reconstruct latent variable to inputs. The encoder and decoder are trained jointly via its loss function, negative Evidence Lower Bound (ELBO) as follows,

$$\mathcal{L}(\theta, \phi; x) = \underbrace{\mathbb{E}_{q_\phi(z|x)} [\log p_\theta(x | z)]}_{\text{Reconstruction Accuracy}} - \underbrace{\beta D_{KL}(q_\phi(z | x) || p(z))}_{\text{Latent Regularization}}, \quad (1)$$

where D_{KL} is the Kullback–Leibler divergence and β controls the KL divergence term: higher β emphasizes the model to focus on mapping the posterior $q_\phi(z|x)$ to input prior $p(z)$, whereas lower β focuses on reconstructing ability given the latent variable z to the original input x . For anomaly detection, loss function elements can be as an anomaly score function Zamanzadeh Darban et al. (2024).

Hundman et al. (2018) use dynamic thresholding to classify anomalies across the healthy cycles. Well known threshold method is to fix a percentile and consider data points above the fixed percentile as anomalies. While this dynamic thresholding does not require meticulous hyperparameter tuning to find the adequate data-suited percentile, the method is useful only when the data contains explicit and obvious anomalous data spikes, since it relies on finding abrupt changes in anomaly score.

2.2 VAEs for CMAPSS

Sanami and Aghdam (2024) adopt Fisher Autoencoder (FAE) (Elkhalil et al., 2021) to detect anomalies in CMAPSS dataset. FAE utilizes Fisher divergence (Ding et al., 2019) to match the shape or geometry of the latent distributions by comparing their gradients. Regardless of the use of Fisher divergence, rather than the conventional KL divergence, no noticeable differences are found when using VAE and FAE.

San Martin and Droguett (2022) evaluate three variants of VAEs, vanilla VAE, LSTM-VAE, and complete sequence VAE, for predicting remaining useful life (RUL). While the work focuses on the supervised RUL prediction task, it provides insightful comparisons between the three VAE variants. They used sliding window approach to feed windowed inputs to the LSTM-VAE which outperformed the vanilla VAE.

Jakubowski et al. (2021) provide extensive experimental results of using VAEs for CMAPSS anomaly detection, achieving substantial improvement compared to other methods. To prevent posterior collapse, β annealing is used in their experiments. However, the biggest gap of their work is that they separated a single set of data into training set and validation set, lacking general performance across other datasets.

3 Method

3.1 CMAPSS Data

CMAPSS dataset provides four subsets which are FD001, FD002, FD003, and FD004. In this work, only the train set will be used since the test set does not contain full trajec-

tory. For training, the dataset is separated into three parts, training (FD001 and FD002), validation (FD003) and test set (FD004).

Additionally, defining anomalies for CMAPSS is required, since there are no explicit anomaly labels inside the data. In this work, any records that have an RUL of fewer than 30 are considered anomalies, otherwise healthy. Further discussions on classifying anomalies using anomaly scores can be found in Section 3.3.

The preprocessing consists of two main steps. First is to calculate RULs by back-tracing the last cycle number up to the first cycle, then label each record either as "Healthy" (0) or "Fatal / Anomaly" (1). Second is to apply min-max scaler for normalizing the data. Normalization is applied only to the 24 input features, and other metadata will remain.

3.2 Time-series VAE

Figure 1a illustrates how the latent variable z is generated given an input x . Encoder q_ϕ has two heads where each outputs μ and $\log \sigma^2$, respectively. Using the reparameterization trick, latent variable z is computed as,

$$z = \mu + \epsilon \odot \sigma,$$

where $\sigma = \exp(\frac{1}{2} \log \sigma^2)$ and $\epsilon \sim \mathcal{N}(0, I)$. Decoder p_θ reconstructs the input $\hat{x}$ given the latent variable z , such that, $\hat{x} = p_\theta(x|z)$.

Figure 1b shows a simple VAE which will be used as a initial baseline, in this work. Whereas Figure 1c shows a GRU-VAE where it leverages GRU layer on top of the baseline model, enabling time-series data. Motivated by San Martin and Droguett (2022), we explore GRU-VAE as a temporal extension of vanilla VAE, where the decoder's output is repeated to match the GRU's input dimension.

3.3 Anomaly Detection

The major challenge of detecting anomalies in CMAPSS dataset is that the anomalies are not explicit, rather gradual. Due to this, two main components for anomaly detection need to be defined, where the first is to define anomaly score and the second is to set the threshold. Anomaly score is a scoring function which measures how anomalous the data point is, and it requires the threshold to classify which points are to be considered as anomalies or healthy points.

In this work, three anomaly scoring functions are considered, each derived from the β -ELBO loss function, which is the most common approach (Zamanzadeh Darban et al., 2024). As shown in Equation 1, three scoring functions are defined, corresponding to the reconstruction term, the KL divergence term, and the full ELBO, as follows:

$$S_{\text{recon}}(x) = \frac{1}{N} \sum_{i=1}^N (x_i - \hat{x}_i)^2 \quad (2)$$

$$S_{\text{KL}}(x) = D_{\text{KL}}(q_\phi(z|x) \| p(z)) \quad (3)$$

$$S_{\text{ELBO}}(x) = S_{\text{recon}}(x) + \beta S_{\text{KL}}(x), \quad (4)$$

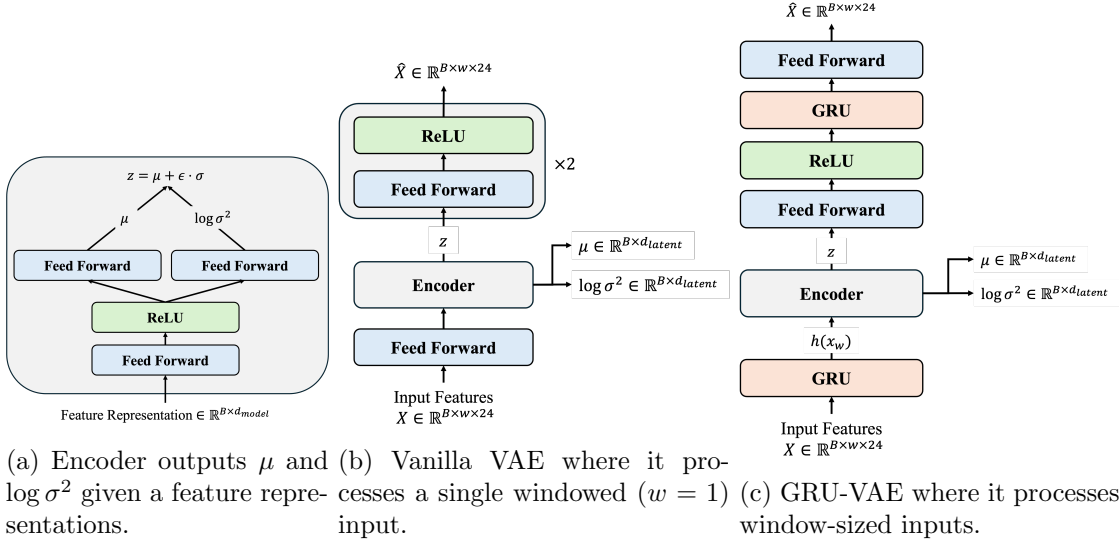


Figure 1: Overall model architecture of vanilla VAE and GRU-VAE. Two models differ by the window size w , where vanilla VAE only can handle single input and GRU-VAE can handle time-series input.

where β is the key element which balances between reconstruction ability and latent mapping ability. Since each anomaly scoring function emphasizes different aspects of the model’s performance, we investigate which scoring function suits best for CMAPSS anomaly detection. Note that if the β is too high, the model pushes the input data point to $\mathcal{N}(0, I)$, no matter what input is fed, eventually causing posterior collapse. The effects of β on anomaly detection performance is further investigated in Section 4.1.

Given the scoring functions, it is essential to find an adequate threshold to discriminate anomalies and other data points. While dynamic thresholding method (Hundman et al., 2018) can avoid fine-tuning, the method is only available when the anomalous data points show sudden changes in values, which is not the case for our problem. Hence, a fixed percentile threshold is employed, calibrated directly from the training set. To be specific, after each training epoch, all training samples are passed through the model and the p -th percentile of the resulting KL scores is computed as the threshold λ . At inference, a data point is classified as an anomaly if its anomaly score exceeds λ :

$$\hat{y} = \mathbf{1} [S(x_{\text{inference}}) > \lambda], \quad \text{where } \lambda = \text{Percentile}_p (S(x_{\text{train}})).$$

4 Experimental Results

The experiments take four processes of evaluations, and each process we fix the corresponding variant: (1) anomaly score comparison, (2) temporal modeling comparison differing w , (3) β sweeps, and (4) threshold sweeps. Each choice is evaluated in isolation by fixing the other at their best-performing values, following an ablation-style experiment. Results are reported on FD003 (validation) and FD004 (test) using AUROC.

	$\beta = 1$		$\beta = 0.05$		$\beta = 0.001$	
$S(x)$	FD003	FD004	FD003	FD004	FD003	FD004
S_{ELBO}	0.53 ± 0.01	<u>0.50</u>	0.63 ± 0.06	0.57	0.83 ± 0.04	0.77
S_{recon}	0.52 ± 0.01	0.51	0.60 ± 0.05	0.58	0.81 ± 0.04	0.82
S_{KL}	0.87 ± 0.12	<u>0.50</u>	0.91 ± 0.02	0.52	0.93 ± 0.01	0.59

Table 1: Anomaly score function comparison. **Bold** numbers represent the highest performance and underline represents the second-best performance.

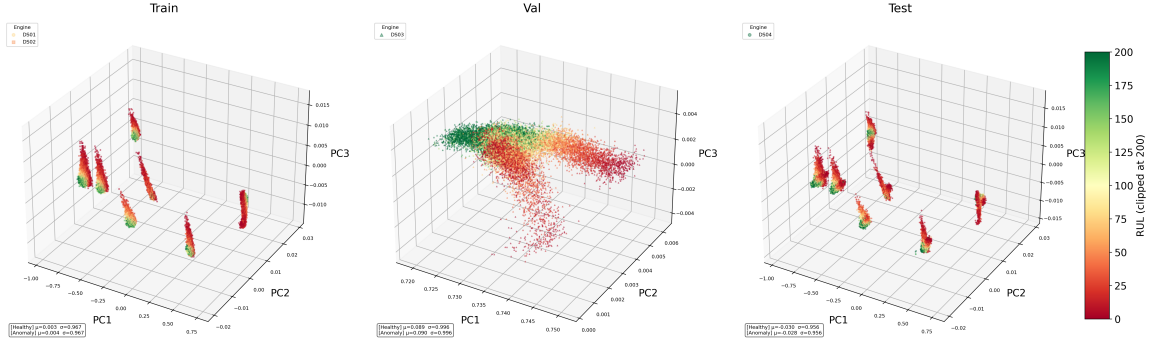


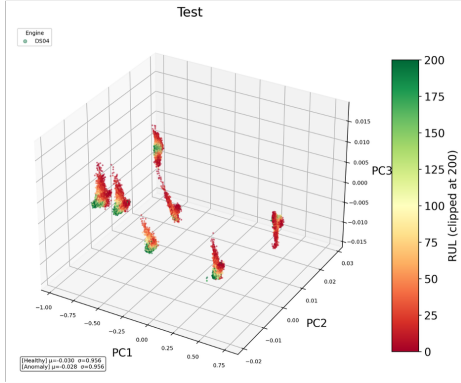
Figure 2: VAE’s latent mapping ($\beta = 1$) by RUL labels. Each set shows the corresponding number of conditions (clusters), such that 7 clusters for training set, 1 cluster for validation set, and 6 clusters for test set.

4.1 Anomaly Score Comparison

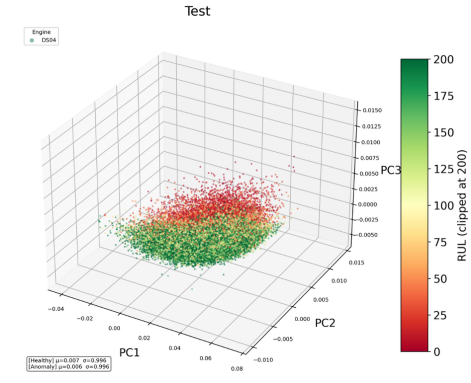
In this section, we investigate which anomaly scoring function leads to the best performance when using vanilla VAE, while fixing other variants. Without losing the generality, β is set to 1, 0.05, and 0.001. In order to measure the performance of detecting anomalies, AUROC score is utilized to measure how coherent the anomaly scores with the actual anomaly labels.

Table 1 presents overall AUROC across three anomaly score functions and three values of $\beta \in \{0.001, 0.05, 1\}$ on validation set (FD003) and test (FD004) set. Results from validation set contain epoch-wise AUROC, therefore it is denoted with mean AUROC with standard deviation, and results from the test set are reported as a single AUROC value computed on the full FD004 evaluation set. S_{KL} consistently achieves the highest validation AUROC regardless of β , suggesting strong discriminative function for FD003. However, S_{KL} fails to generalize for FD004 subset, where it performs near random across all β s (0.50 – 0.59). In contrast, S_{recon} shows the highest AUROC for FD004 across all β values. S_{ELBO} shows the intermediate performance of S_{recon} and S_{KL} , as it simply is a sum of the two.

The performance gap within different score functions is attributed to the data distribution difference between FD003 and FD004. Figure 2 illustrates the latent representations of VAE ($\beta = 1$), using principal component analysis (PCA). While all sets show clear degradation pattern, FD003 and FD004 differ in how the latent space is structured. For FD003, the latent variables form a single cluster where the distance from the prior $\mathcal{N}(0, I)$ correlates with health state, making S_{KL} an informative anomaly signal. In contrast, FD004’s latent space is partitioned by operating conditions rather than health state, so the distances



(a) Latent space of VAE ($w = 1$, $\beta = 1$). VAE encodes inputs to clustered latent variables.



(b) Latent space of GRU-VAE ($w = 10$, $\beta = 1$). GRU-VAE encodes inputs to a single cluster.

Figure 3: Latent space comparison between VAE and GRU-VAE.

between the data points reflect condition differences rather than degradation, leading S_{KL} uninformative.

In summary, no single scoring function dominates across all settings, as β affects overall performance. S_{KL} achieves the highest performance when the latent space forms a well-structured single cluster, as observed from FD003, while S_{recon} outperforms when latent variables are separated into multiple clusters, as FD004. This sensitivity to latent structure motivates the next experiment, which investigates whether architectural choices can produce more reliable latent representations.

4.2 Effects of Window Sizes to VAEs

Here, we compare VAE ($w = 1$) and GRU-VAE ($w = 10$) by evaluating their latent space structure on test set (FD004) at $\beta = 1$ considering β -sensitivity. Previously shown, S_{KL} achieved the most discriminative score function when the latent space is well-structured, making the latent space quality a critical determinant for anomaly detection. However, VAE with $\beta = 1$ showed a distributed clusters by operating conditions, and simply increasing β to get an single cluster could potentially cause posterior collapse. We investigate whether GRU-VAE’s temporal encoding suppresses operating condition regions in latent space. Note that AUROC is not measured here, as the two architectures respond differently to the same β , making direct performance comparison misleading. Further fine-tuning of β is discussed in the next section.

Figure 3 demonstrates the latent spaces of VAE and GRU-VAE with the same $\beta = 1$. As shown, VAE produces six distinct clusters corresponding to FD004’s operating conditions, while GRU-VAE produces a single compact latent cluster. This behaviour is due to the model’s architectural difference between the two encoders. VAE processes each cycle by encoding the dominant feature of the input, the operating condition, causing the encoder to map the inputs per operating conditions. Whereas GRU-VAE processes inputs per 10-cycles that GRU layer compresses temporal differences within the window, rather than the cycle

	$\beta = 0.001$		$\beta = 0.1$		$\beta = 10$	
$S(x)$	FD003	FD004	FD003	FD004	FD003	FD004
S_{ELBO}	0.81 ± 0.01	0.52	0.66 ± 0.07	0.52	0.76 ± 0.08	<u>0.49</u>
S_{recon}	0.81 ± 0.01	0.52	0.64 ± 0.07	0.52	0.76 ± 0.08	<u>0.49</u>
S_{KL}	0.41 ± 0.10	<u>0.49</u>	0.88 ± 0.02	<u>0.49</u>	0.68 ± 0.22	0.94

Table 2: β sensitivity analysis of GRU-VAE ($w = 10$) across scoring functions.

summary. Since operating condition primarily affects values from sensors while degradation is considered as gradual temporal drift, GRU-VAE’s latent space captures health-relevant temporal patterns. Nevertheless, GRU-VAE shows a risk in posterior collapse if β is not tuned, since both the healthy and anomalous latent variables are pushing toward $\mathcal{N}(0, I)$. Given this, we fix the window size $w = 10$ and further expand the β tuning of GRU-VAE for achieving higher performance without posterior collapse.

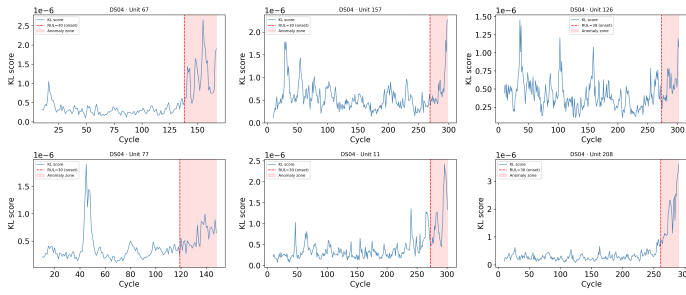
4.3 β Tuning

In this experiment, we explore AUROC performance differences using three score functions (S_{ELBO} , S_{recon} , and S_{KL}) for a fair comparison with varying $\beta \in \{0.001, 0.1, 10\}$ for GRU-VAE model ($w = 10$), as shown in Table 2. For the validation set (FD003), the highest AUROC across β values hover from 76% to 88%, which is expected, as the validation set contains a single operating condition, making the anomaly scores more discriminative. When it comes to the test set (FD004), AUROC scores range from 49% to 52%, except for one case where $\beta = 10$ and $S_{\text{KL}}(x_{\text{test}}) = 0.94$. The posterior of the GRU-VAE with $\beta = 10$ collapsed during the training phase, yet achieves the highest performance. This contradicts with a norm where posterior collapse is considered to be a failure of the model.

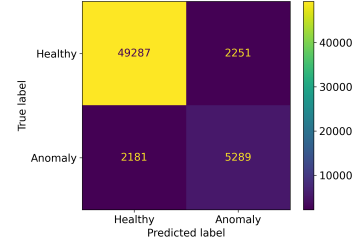
The main reason for its discriminative anomaly scores with KL term is driven by the encoder’s behaviour. By having a high $\beta = 10$, the model penalizes the encoder to push all healthy training inputs tightly onto $\mathcal{N}(0, I)$, resulting in near-zero KL losses. However, when unseen anomalous samples are fed to the model, the encoder fails to map anomalous samples tightly onto $\mathcal{N}(0, I)$, since the anomaly data samples are out-of-distribution. This failure mapping produces non-zero KL scores for anomalies which differentiates them from the near-zero KL scores from healthy samples. Even when the absolute KL score difference between the healthy samples and anomalies is small, the consistent ranking of anomalous samples above healthy ones is sufficient to yield a high AUROC.

4.4 Fixed Percentile Alert Threshold

We evaluate the GRU-VAE’s anomaly detection performance by confusion matrix using the fixed percentile alert threshold. As a consequence of posterior collapse, training KL scores are uniformly near-zero, making the percentile threshold insensitive to p , as all percentile values produce equivalent classification boundaries (evaluations are done with $p \in \{70, 85, 95, 99\}$, but the results are same as the following results). Thus, percentile is fixed $p = 70$, and rather we focus on the overall performance. This experiment intentionally collapses the model’s posterior to achieve discriminative S_{KL} by setting $\beta = 10$.



(a) KL score over degradation trajectories. Y-axis is scaled by $1e-6$.



(b) F1 score: 0.705, precision: 0.701, recall: 0.708

Figure 4: Performance overview of GRU-VAE ($w = 10$, $\beta = 10$).

Figure 4a demonstrates the KL score differences between healthy and anomaly samples within a full degradation trajectory, which shows six randomly sampled trajectories from the test set. Unit 67, 11, and 208 especially illustrates how the KL scores of anomalies differ by healthy samples. Given these differences in KL scores across healthy and anomalies, Figure 4b shows the confusion matrix, with a F1 score of 0.7. As described in Section 4.3, encoder mapped healthy samples strictly to $\mathcal{N}(0, I)$, leading to near-zero KL scores. However, when the unseen anomalous data points are given, encoder failed to map the samples to $\mathcal{N}(0, I)$ as close as the healthy samples, resulting in relatively higher KL scores. Further detailed analysis is covered in Appendix B.

5 Conclusion

This work empirically evaluated β -VAE-based anomaly detection on CMAPSS dataset, exploring anomaly score functions, window size impacts, effects of β , and the model performance using fixed percentile threshold method. Experiments suggest that using a KL loss as an anomaly score function yields the highest AUROC performance when the latent space contains an organized cluster, since the KL can differentiate between the healthy and anomaly samples. While using a reconstruction loss as a score function generalizes more reliably to more complex, clustered latent space when the β is well-tuned. Notably, posterior collapse unintentionally resolves the sparse latent space issue by pushing all data points to $\mathcal{N}(0, I)$, classifying anomalies with KL loss effectively. Nevertheless, the performance gap between FD003 and FD004 indicates that the problem has not yet been solved.

Current work compares various β -VAE configurations, but it lacks generalization, as it requires extensive hyperparameter tuning and decision for which metric to be used for score function and for alert thresholding method. Future work includes controlling β , such as β annealing or free bits, to prevent posterior collapse while leveraging reconstruction loss as the score function. Additionally, adaptive thresholding method should be explored, since current work adopts a fixed percentile threshold of $p = 70$, enabling more robust performance.

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Appendix A. Additional CMAPSS Data Information

CMAPSS dataset provides train set and test set, each set consists of FD001, FD002, FD003, and FD004, each derived from different types of engines. Because each subset is composed of different condition(s) and fault mode(s), the difficulties also vary by the set. Generally, FD004 is considered to be the hardest subset since it contains numerous trajectories, with complex conditions and modes. Table 3 shows overall structure of the dataset.

Dataset	# of Trajectories	Condition(s)	Fault Mode(s)
FD001	100	1 (Sea Level)	1 (HPC Degradation)
FD002	260	6	1 (HPC Degradation)
FD003	100	1 (Sea Level)	2 (HPC Degradation, Fan Degradation)
FD004	249	6	2 (HPC Degradation, Fan Degradation)

Table 3: Overview of CMAPSS

In the dataset, each record provides 3 operational settings and 21 sensor data, which will serve as an input. Other fields, such as unit, engine, and RUL, are used as metadata for inference stage. During inference, the model’s performance will be measured per engine-unit pair and aggregated, since the data distributions vary by the engines.

Appendix B. Performance Analysis

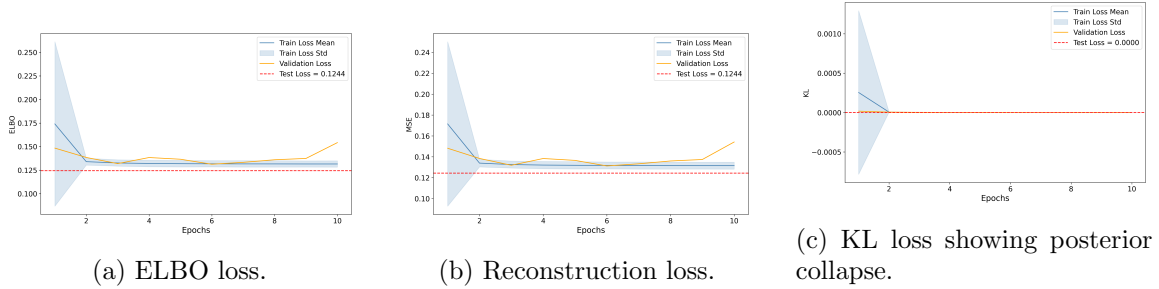
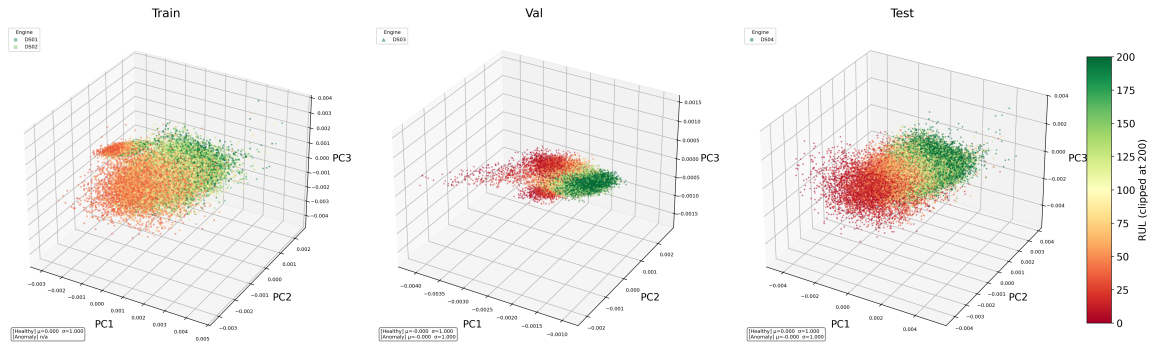


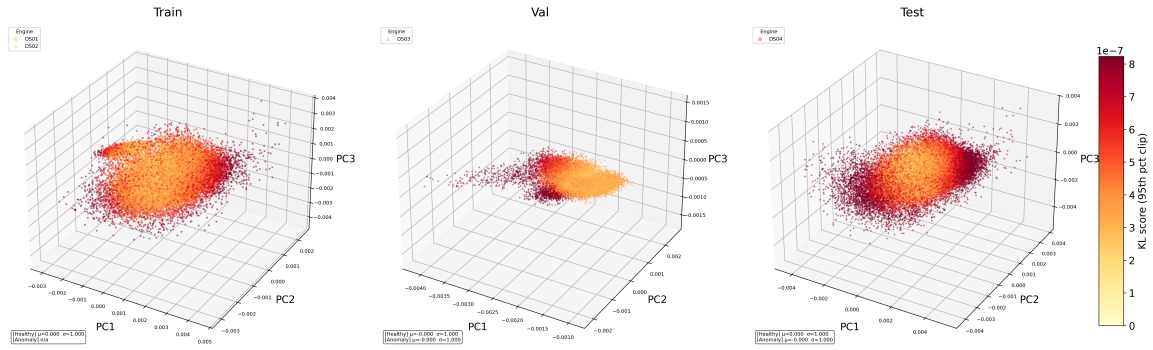
Figure 5: Training result of GRU-VAE ($w = 10$, $\beta = 10$) over 10 epochs.

Figure 5 shows the training result of GRU-VAE ($w = 10$, $\beta = 10$) from Section 4.4. As expected, KL loss converges to near-zero values, thus the reconstruction loss resembles a lot with ELBO loss. The results show that the model is mapping all inputs, whether healthy or anomalous, to $\mathcal{N}(0, I)$, which in result, produces near-zero KL loss.

Figure 6 visualizes the latent space representations by RULs and KL scores. As previously, latent mapping is tightly bounded to $\mathcal{N}(0, I)$, forming a well-structured cluster, while posterior collapsed.



(a) Latent space of GRU-VAE ($w = 10$, $\beta = 10$) by RULs.



(b) Latent space of GRU-VAE ($w = 10$, $\beta = 10$) by KL scores.

Figure 6: Latent space representations difference across KL scores and its RULs indicating healthy or anomaly data points.